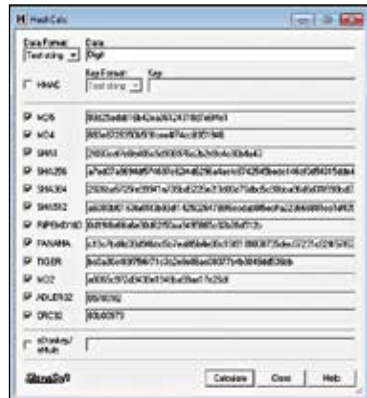


the key in order to break the cipher. In most cases, the only way to figure out the key would be to run a brute-force attack which would use the algorithm with every possible input as a key. Once the data is decrypted in an acceptable pattern, it can stop and record (and display) the key that was used for encryption. Now, when it comes to brute-forcing a cryptographic algorithm, it's difficult to do it for a simple reason - most algorithms are computationally intensive and require more computational power. To break them would take plenty of computational resources and time. That brings us to another factor - time.

Time

We mentioned that your mobile phone can't crack DES or RSA algorithms, but it in fact can crack any algorithm. The only problem - it will perhaps take millions of years to break it! That would be pretty impractical, wouldn't it?

Apart from requiring a strong algorithm and huge amounts of computational powers, it would take time to break a cipher. However we have an interesting case - DES. DES was an algorithm which was developed in the 1970s and was so popular that it became a standard. The algorithm's full name happens to be 'Data Encryption Standard'. When it was invented, it was said to be very strong. Computers in those days would have taken eons to break it. Today, many loopholes have been discovered in the algorithm and it has been proven vulnerable. Time is an interesting factor when it comes to cryptography however, in most cases, the data protected by an algorithm which is deemed to be strong in its own time becomes useless by the time the cipher becomes vulnerable due to further research in the area or due to increase in computing capability.



HashCalc is a good tool with small footprint to generate hashes for popular algorithms

Encryption keys

RSA is widely known for being a difficult nut to crack! This is because RSA is an algorithm that depends on the difficulty of finding the factors of the product of two prime numbers. The product of two prime numbers